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Functional Requirements Document (FRD)

Automated Data Pipeline for
Client Reporting

Project Title: Automated Data Pipeline for Client Reporting

Date:

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Project Overview:

This project aims to design and implement an automated, cloud-native data pipeline to ingest, transform, and load structured business data from source systems (S3) into Amazon Redshift for downstream analytics and reporting. The pipeline will support daily data refreshes, error handling, and monitoring, enabling accurate and timely decision-making.

Business Requirements:

- ✓ Automate the ETL workflow to reduce manual dependencies and data latency.
- ✓ Provide reliable, daily-updated data to business intelligence tools for reporting.
- ✓ Enable data validation and transformation before loading into Redshift.
- ✓ Ensure compliance with security, auditing, and performance standards.

User Roles & Permissions:

Role	Access Level
Data Engineer	Full access to ETL code, logs, and pipeline monitoring
Business Analyst	Read-only access to Redshift tables, error logs, and reports
Admin	IAM access management, manual job trigger rights
Client (Optional)	Access to daily report outputs and status summaries via email or portal

Functional Requirements:

ID	Description	Priority	Source
FR-01	System shall fetch CSV/JSON files from Amazon S3 daily.	High	Business Team
FR-02	AWS Glue shall transform raw data based on business logic (e.g., date format standardization, null removal).	High	Data Team
FR-03	Transformed data shall be loaded into staging tables in Redshift.	High	Tech Lead
FR-04	A Lambda function shall validate the file structure and schema before ETL begins.	Medium	QA Analyst
FR-05	System shall log all job execution status and errors in CloudWatch.	High	DevOps Team
FR-06	SNS shall notify stakeholders via email upon success/failure of the pipeline.	Medium	Stakeholders
FR-07	Users shall be able to trigger manual runs via a web interface or API.	Low	Product Owner

Data Flow Overview:

Source: Raw data in Amazon S3

- **Transformation:** AWS Glue Jobs (ETL logic).
- **Validation:** Schema checks via Lambda.
- **Load:** Amazon Redshift staging & reporting tables.
- **Notification & Logs:** CloudWatch + SNS email alerts.

Dependencies:

- ✓ Amazon S3 for data storage.
- ✓ AWS Glue for ETL processing.
- ✓ Redshift for analytics layer.
- ✓ Lambda for validation logic.
- ✓ IAM roles and security policies.
- ✓ CloudWatch & SNS for monitoring and alerting.